

Element F

Design Viability Description	Connection to Design Requirement	Credible Evidence
The final prototype is cost-friendly	Final prototype does not cost more than \$20	Materials used: - Blinds \$10 - Spray Paint \$5 - Hinges \$3 Total Cost = \$18
The design includes a push-button to open and close the blinds based on the timer, which would function at certain times of the day.	Final Prototype should be programmable	A Push Button was drilled onto the side-panels of the box. A wire was connected from the push-button to the timer, so information can travel between both ways.
The design uses light-weight materials, so as not to cause the user inconvenience	Final Design should be light-weight	Weight of Materials - Wood 2 lbs - Motor 2 lbs - Blinds 1 lbs Total Weight = 5 lbs
The prototype uses light-weight materials, which allows the design to be portable by users without causing any inconvenience.	Prototype should be portable	At the design-a-thon, in order to test this requirement, we asked users to lift the box from Point A and place it at Point B. Through their feedback, they said they didn't experience any inconvenience in doing so.
The design is programmable through a motor and a timer.	Prototype operates on its own using a source that is environmentally friendly	The timer is set to go off at a certain time of day depending on the plant in the box, and the timer gives a signal to the motor to turn on; which opens and closes the blinds.
The design was built from sturdy materials such as strong wood, but all the other smaller parts were made from plastic. This way, the final design wouldn't be more than 20lbs.	Design should support 10lbs of weight.	To test this requirement, we placed weights on the prototype, starting little by little. Our prototype was able to hold 20 lbs of weight.